

FORTALEZA, Brazil

WMO: 823980

Lat: 3.776S

Lon: 38.533W

Elev: 25

StdP: 101.03

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	23.0	16.1	11.5	28.8	17.2	12.3	28.6	10.4	29.0	10.0	28.9	2.9	180	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.1	32.1	24.8	31.8	24.7	31.2	24.5	26.6	29.3	26.2	29.1	26.0	29.0	6.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	28.0	25.2	20.4	27.5	25.1	20.3	27.4	83.3	29.5	81.6	29.3	80.3	29.4	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.4	8.6	DB	21.4	32.8	0.9	1.0	20.8	33.5	20.2	34.1	19.7	34.6	19.0	35.3	(4)
(5)			WB	18.4	27.7	1.0	1.4	17.7	28.7	17.1	29.5	16.6	30.3	15.9	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	27.9	27.8	27.5	27.3	27.5	27.1	27.0	27.1	27.5	27.8	28.1	28.3
	DBStd	0.99	1.13	1.22	1.16	1.02	0.87	0.87	0.83	0.72	0.70	0.67	0.65	0.76
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6410	554	497	542	519	541	513	526	530	524	553	542	567
	CDD18.3	3368	296	264	284	269	283	263	267	272	274	294	292	309
	CDH23.3	34735	3203	2845	2953	2723	2918	2578	2531	2635	2737	3053	3144	3414
	CDH26.7	10016	976	886	884	782	859	699	634	678	734	864	941	1079
Wind	WSAvg	4.6	4.3	3.8	3.3	3.2	3.6	4.1	4.6	5.4	5.9	5.9	5.6	5.1
Precipitation	PrecAvg	1589	136	198	341	351	231	146	83	26	18	13	12	41
	PrecMax	2841	668	504	831	668	670	568	298	190	147	56	47	210
	PrecMin	890	10	10	73	57	44	16	0	0	0	0	1	2
	PrecStd	502	120	125	147	157	125	110	69	32	24	14	11	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.2	32.1	32.0	31.9	31.8	31.2	31.7	31.8	31.9	32.0	32.2
		MCWB	25.2	25.5	25.2	25.3	25.1	24.4	23.2	23.0	22.9	23.6	24.2	25.0
	2%	DB	31.5	31.9	31.3	31.2	31.1	31.0	30.9	31.0	31.1	31.1	31.2	31.8
		MCWB	25.0	25.3	25.3	25.2	25.0	24.2	23.3	22.9	23.1	23.5	24.1	24.9
	5%	DB	31.0	31.1	31.0	30.9	30.8	30.2	30.2	30.2	30.5	30.9	31.0	31.1
		MCWB	24.8	25.1	25.3	25.2	25.0	24.0	23.2	22.8	23.0	23.5	24.1	24.6
	10%	DB	30.2	30.3	30.1	30.1	30.1	29.9	29.8	29.9	30.0	30.1	30.2	30.6
		MCWB	24.6	25.1	25.2	25.1	24.8	24.0	23.1	22.8	22.9	23.3	23.8	24.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.3	27.4	27.2	26.7	26.5	26.0	25.4	25.5	25.2	25.4	25.7	26.2
		MCDB	29.8	29.9	29.3	29.4	29.5	29.2	28.1	28.2	28.5	29.2	29.6	30.1
	2%	WB	25.9	26.6	26.4	26.2	26.1	25.5	25.0	24.8	24.7	24.9	25.1	25.7
		MCDB	29.4	29.2	29.0	29.0	29.1	28.6	27.7	27.7	28.0	28.9	29.2	29.5
	5%	WB	25.6	26.2	26.1	25.9	25.7	25.1	24.6	24.4	24.2	24.5	24.8	25.2
		MCDB	29.0	28.9	28.9	28.7	28.7	28.2	27.5	27.5	27.7	28.4	28.9	29.0
	10%	WB	25.2	25.8	25.7	25.7	25.5	24.9	24.2	24.1	24.0	24.2	24.5	25.0
		MCDB	28.5	28.7	28.6	28.4	28.4	27.9	27.2	27.3	27.6	28.0	28.6	28.8
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.4	5.5	5.6	5.8	6.0	6.2	6.3	5.9	5.5	5.3	5.1
		MCDBR	5.6	5.8	6.0	6.3	6.3	6.4	6.6	6.6	6.2	5.8	5.5	5.4
		MCWBR	1.8	2.1	2.2	2.4	2.3	2.4	2.7	2.8	2.4	2.1	1.8	1.7
	5% WB	MCDBR	5.4	5.6	5.7	5.9	5.9	6.1	6.2	6.3	5.9	5.5	5.4	5.3
MCWBR	2.2	2.5	2.6	2.5	2.4	2.4	2.6	2.7	2.4	2.2	1.9	1.9	1.9	
Clear-Sky Solar Irradiance	taub	0.418	0.414	0.414	0.406	0.405	0.405	0.391	0.407	0.438	0.431	0.421	0.426	
	taud	2.456	2.479	2.489	2.530	2.517	2.509	2.545	2.468	2.350	2.366	2.405	2.411	
	Ebn at Noon	918	921	910	894	869	855	875	881	876	894	908	906	
	Edh at Noon	118	117	114	106	103	102	99	111	129	129	124	123	
All-Sky Solar Radiation	RadAvg	5.57	5.42	5.25	4.88	5.13	5.10	5.45	6.15	6.53	6.58	6.44	6.00	
	RadStd	0.44	0.46	0.46	0.45	0.38	0.29	0.21	0.18	0.20	0.14	0.17	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page